ALAN BARNES

(630) 501-6564 • Hanover Park, IL
alanbarnes447@gmail.com
www.linkedin.com/in/alanrbarnes
https://alanrbarnes.com

SOFTWARE DEVELOPER

Superior Computer Programming Skills That Drive Efficiency and Bottom-Line Results

A self-taught software engineer who spent 20 years in manufacturing and CNC programming before teaching myself C# and JavaScript and earning a B.S. in Computer Science while working full-time. Today I build production web applications, RESTful APIs, and cloud infrastructure with the same discipline and efficiency I learned on the factory floor.

TECHNICAL SKILLS AND AFFILIATIONS

Languages: C#, TypeScript, JavaScript, Python, Java, SQL, HTML, CSS

.NET / Backend: ASP.NET, ASP.NET Core, Entity Framework, RESTful API Design

Frontend: Angular, React, Bootstrap, Tailwind CSS

Databases: SQL Server, MySQL, MongoDB

Cloud & DevOps: Azure Cloud, AWS Cloud, IBM Cloud, Docker, IIS, Git, Postman API Platform

Tools & Environments: Microsoft Visual Studio, Visual Studio Code, Windows, Linux, Mac OS X

Additional Platform Experience: PHP, Laravel, Node.js, Express, Electron, WordPress, Umbraco, Joomla, Ektron, Moodle

Affiliations: Institute for Electrical and Electronics Engineers, Association for Computing Machinery

PROFESSIONAL EXPERIENCE

Senior Level Programmer/ Web Developer, Living Word Christian Center, Forest Park, IL 2021 – present
Globally televised Christian Church, business education leader, and publisher

Snapshot: Responsible for full-stack web and mobile application development, ecommerce, and server/cloud/database administration for a multi-site media and ministry organization, with a focus on automating manual processes to reduce labor costs.

- Administration of cloud infrastructure across Azure, AWS, and IBM Cloud including cloud computing, storage, CDN, and networking, along with cloud security (WAF, firewalls, IAM, encryption, SSL) and database management systems (MySQL, SQL Server).
- Migrated and consolidated infrastructure including 3 legacy Rackspace dedicated windows servers, multiple Azure virtual Machines and scattered shared/PHP hosting accounts onto 2 modernized, actively maintained windows servers and one dedicated Linux server, cutting hosting costs by $3,000/month (~$36,000 annually) and reducing ongoing maintenance.
- Administration of database management systems (DBMS) such as MySQL and SQL Server.
- Maintain and support 30+ websites across 100+ domains for a dual-purpose media/publishing and ministry organization, including 3 ASP.NET Core/Angular applications, 5 Ektron (classic ASP.NET) applications, 1 Umbraco site, and numerous WordPress/HTML sites.
- Automate data processing and reporting tasks using Python-based CSV scripts.
- Built and maintained RESTful APIs powering 3 ASP.NET Core/Angular applications, and worked with GraphQL endpoints connecting WordPress-based content to front-end applications.
- Implemented performance optimization techniques to improve website loading times and user experience.

CNC Programmer and Production Supervisor, Chicago Mold, St Charles, IL 2000 – 2021
Leading manufacturer of plastic injection and compression molds for thermoset and thermoplastic industries

Mold Maker (2000 – 2003)
CNC Programmer and Production Supervisor (2003 – 2021)

Snapshot: Responsible for analyzing job orders and simplifying machine tool processes to increase productivity for clients. Created innovative solutions using C#, JavaScript, script macros, and post processors to automate manufacturing workflows. Directly managed two employees while overseeing a staff of 75.

- Technical problem-solver; use varied computer languages including Java Script to automate tedious and time-consuming tasks.
- Created application in C# used to interface with the client side of the cam application to include special functionality and automate redundant tasks.
- Created post processor used with cam software to automate machines; wrote numerous script-based macros to increase productivity.
- Designed algorithm-based method for picking up rotational centers on parts by synchronizing computer with the machine; eliminated the need to buy a $20,000 probing system.
- Develop department process improvements including eliminating antiquated machines and upgrading to automated machines that do not require operator oversight, work at higher speed, and run overnight; increased production capabilities by 75%.

"Alan is someone I can give a job to and he will figure out how to do it no matter what." – Owner, Chicago Mold

Workgroup Member, Institute for Electric and Electronic Engineers, Chicago**,** IL 2019 – 2021
Leading developer of industry standards on a broad range of technologies

Snapshot: Collaborate with stakeholders from IU Health, Medtronic, Draeger Medical Systems, and Columbia University to develop standards for IOT device manufacturers to ensure that new medical devices can connect to the internet securely and to limit unauthorized access.

- Researched existing privacy and security frameworks used currently in conjunction with the internet and internet related devices to help inform protocols to incorporate greater privacy.

EDUCATION

Bachelor of Science (B.S.), Computer Science, University of Illinois, Springfield, IL December 2020

Cumulative GPA: 3.51/4.00

Coursework: Algorithms & Computations, Data Structures and Algorithms, Software Engineering, Web Development & Programming, Web Design, Applied Statistics, Intro to Operating Systems, Intro to Database Systems, Web Design, Programming Languages, Java Programming

Professional Development & Continuing Education:

- Completed 100+ hours of advanced training in C#, Python, AI Engineering, ASP.NET Core, JavaScript, Angular, React, SQL Server, and cloud architecture though technical books, Udemy, and extensive self-directed study.
- Engage in ongoing professional development focused on modern software architecture, design patterns, and secure API development.
- Continuously evaluate emerging technologies, frameworks, and development practices to support modern web application development.